

ELECENG 2EI4

Project 4: CMOS XOR GATE

Vaishnav Jayaraj
jayarajv
400588635

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1. Circuit Schematics & Sizing

1.1 Circuit Schematic

The circuit schematic can be extracted algorithmically from a derivation of the Boolean expression for an XOR gate.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Table 1: Truth table for XOR gate

Boolean Expression for XOR Gate:

$$Y = A \oplus B$$

$$Y = \bar{A}B + A\bar{B}$$

$$Y = \overline{\bar{A}B + A\bar{B}}$$

$$Y = \overline{(\bar{A}B) \cdot (A\bar{B})}$$

$$Y = \overline{(A + \bar{B}) \cdot (\bar{A} + B)}$$

$$Y = \overline{AB + \bar{A}\bar{B}}$$

Once a Boolean expression in the negative form is found, a CMOS circuit can be constructed using pull-down and pull-up networks (PDN & PUN). These two networks will have mirrored properties in construction which will result in the accurate logic to be formed. The PDN consists of NMOS, while the PUN will consist of PMOS. Logic rules can be used to construct the circuit utilizing an inverter CMOS circuit for any inversions to be made.

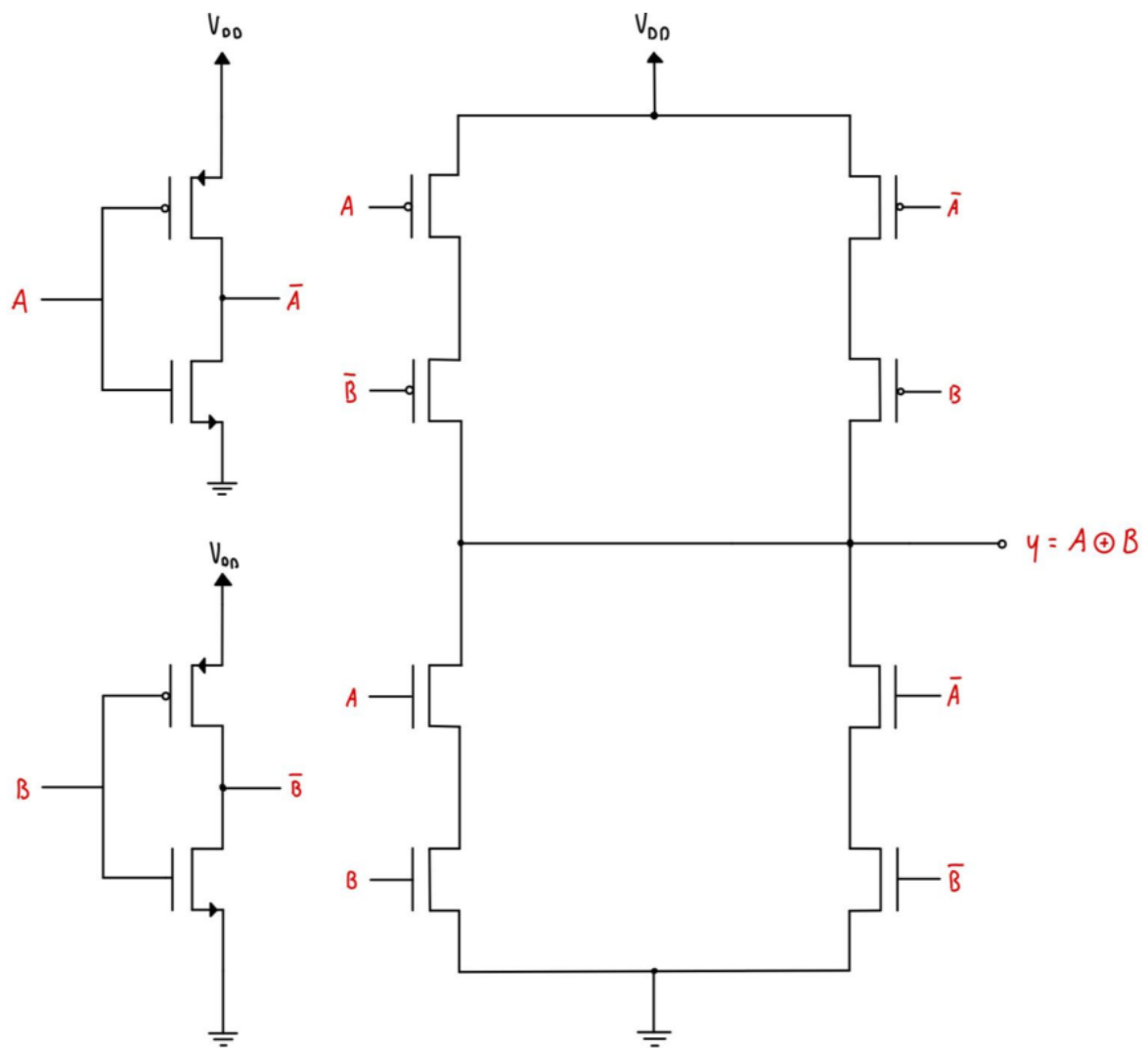


Figure 1: Circuit Schematic of CMOS Inverter and XOR gate

1.2 Ideal Sizing

To ensure optimal performance of the MOSFET based XOR circuit, it is appropriate to appropriately size the PMOS and NMOS transistors. PMOS devices conduct with holes, which are much slower at conducting than the electrons used in NMOS devices. This leads to PMOS having to be physically much wider to match the speed of the NMOS.

The average sizing for the PMOS and NMOS are:

$$PMOS: \left(\frac{W}{L}\right)_P = \frac{5}{1}$$

$$NMOS: \left(\frac{W}{L}\right)_N = \frac{2}{1}$$

Hence it can be demonstrated that the mobility ratio between electrons and holes is around 2.5. This means that NMOS transistors are approximately 2.5X faster than PMOS.

From the CMOS-based XOR network designed, the longest path for both the PUN and PDN network are 2 transistors in series. Therefore the total effective resistance for each network is halved. This leads to half the speed for PDN and PUN. Using this the sizing ratio between the PMOS and NMOS transistor can be calculated to be 2.5 as shown:

$$PMOS: NMOS = \frac{2P}{2N} = \frac{5}{2} = 2.5$$

Hence the PMOS must be approximately 2.5 times larger than the NMOS size in order to achieve maximum propagation delay.

1.3 Feasibility of Ideal Sizing

The circuit schematic has 2 PMOS transistors and 2 NMOS transistors. This path is from VDD to GND meaning that the sizing of the circuit is also $2P/2N = 2.5$, the same value as the ideal sizing. Because of this, the design will be feasible to implement, with its symmetry helping play a role in implementation

2. Testing on AD3

The designed circuit was built on a breadboard:

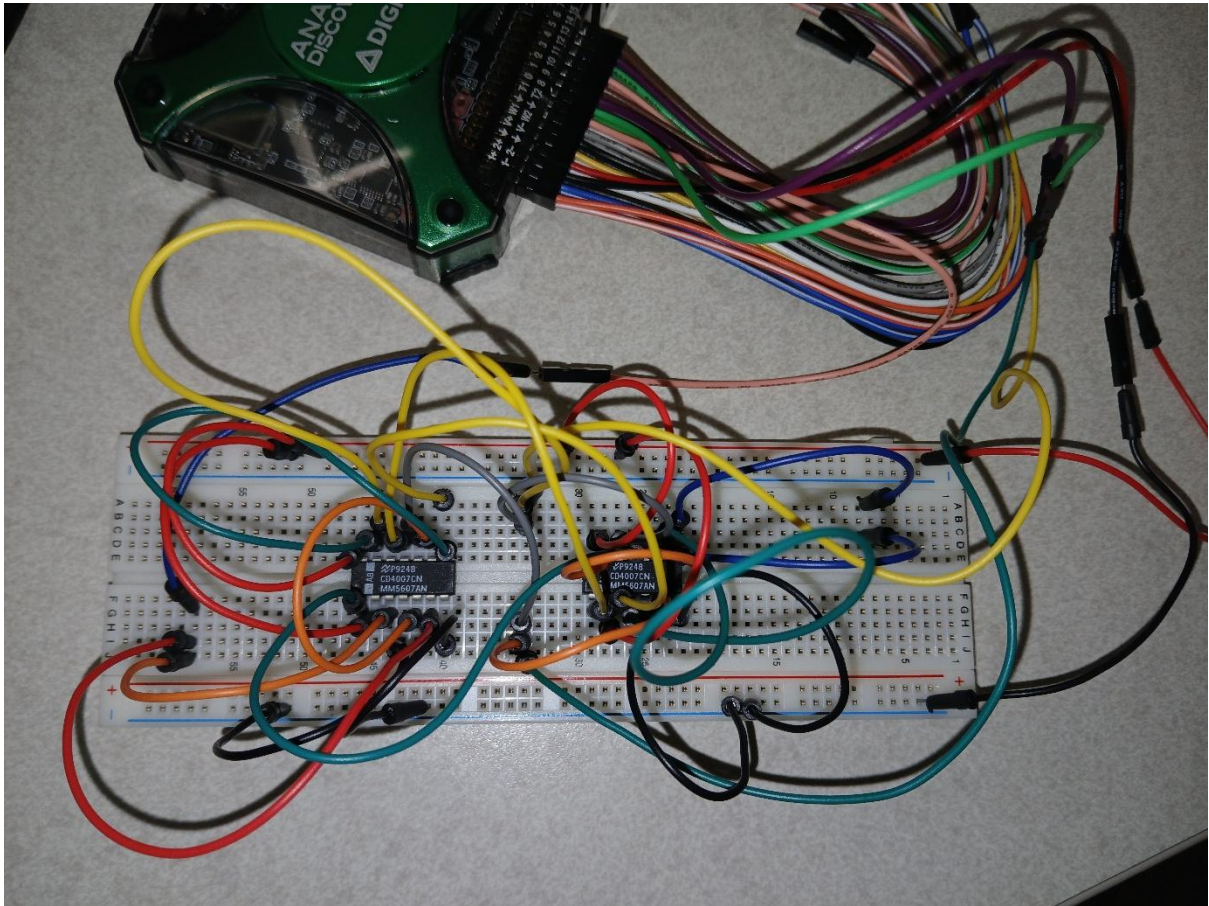


Figure 2: Pic of physical circuit

2.1 Functionality Testing

For this circuit, several digital IO logic probes were tracked using the AD3. The DIO 0, DIO 1, and DIO 2 probes were connected to input A, B, and output Y respectively. A 5V DC supply was then connected to VDD, with a common ground implemented.

The Pattern 1 channel of the AD3 is then configured to set Input A (DIO 0) to a clock signal with 2KHz and Input B (DIO 1) to a clock signal with 1KHz.

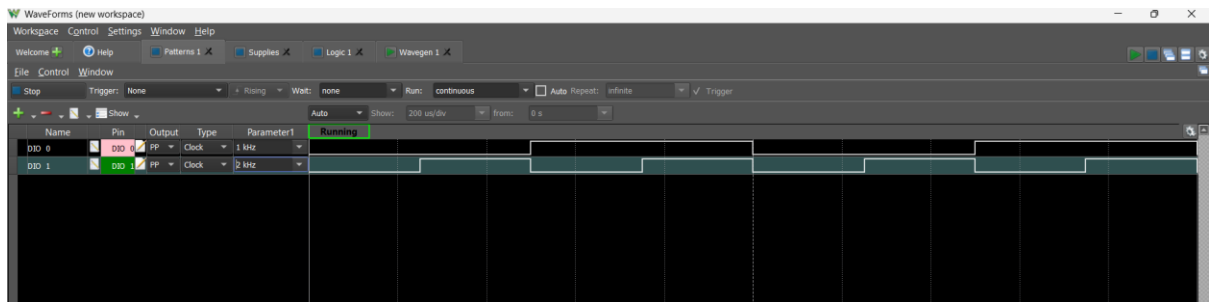


Figure 3: Pattern input signals on AD3

This will ensure that all the required of input combinations can be achieved to recreate the truth table for the XOR Gate.

Finally, the logic 1 channel can be run to analyse the result.

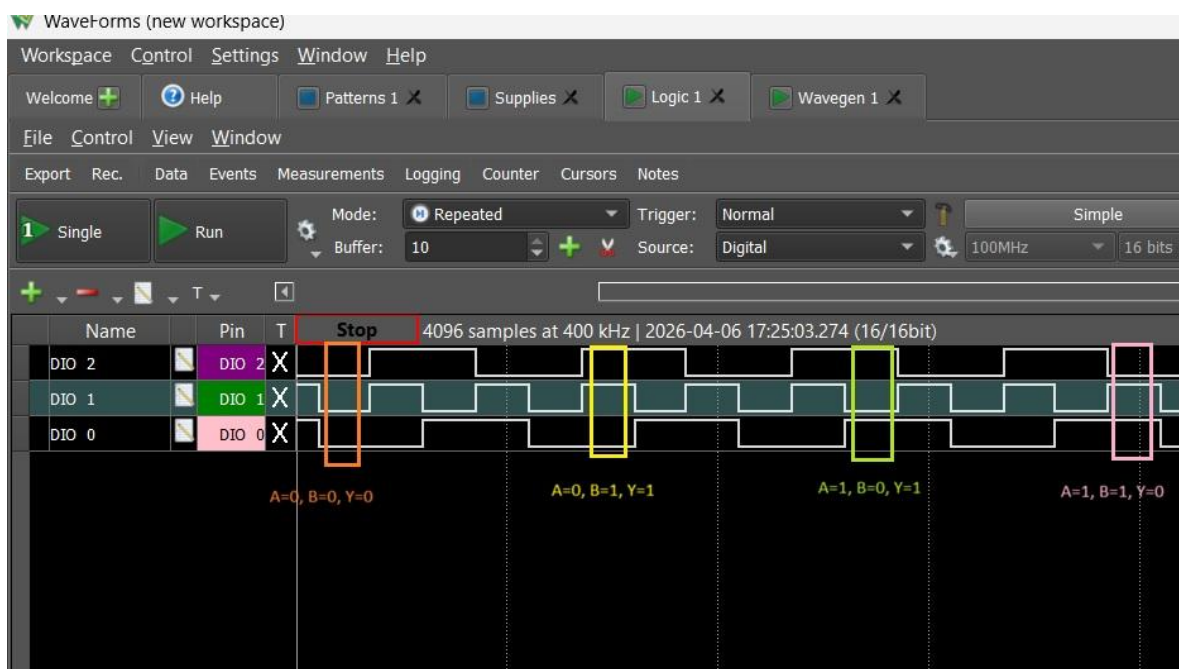


Figure 4: Digital output of both input and output signals

As seen from the figure the circuit successfully mimics an ideal XOR Gate as the output (Y) only switches to 1 whenever there is an odd number of 1's in the input (A&B).

2.3 Static Level Testing

To determine the low and high voltage levels for the circuit, either input was kept at 5V DC with the other having a 5-0V square wave. This allows us to experimentally measure the maximum high and low voltage logic output at Y.

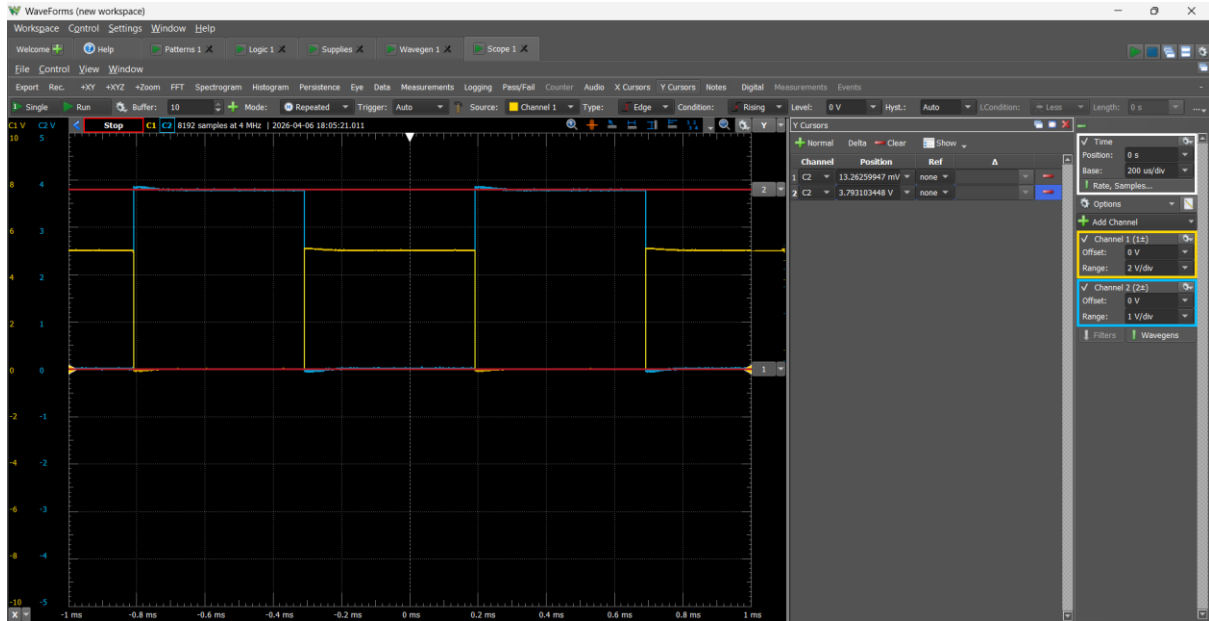


Figure 5: Visualization of logic high and low voltage when input B is fixed and input A is alternating

From the Fig above, when A is alternated between 0-5 V and B is kept high at 5V, the high logic state (V_H) was measured to be 3.79V and the low logic state (V_L) to be approximately 13.27 mV.

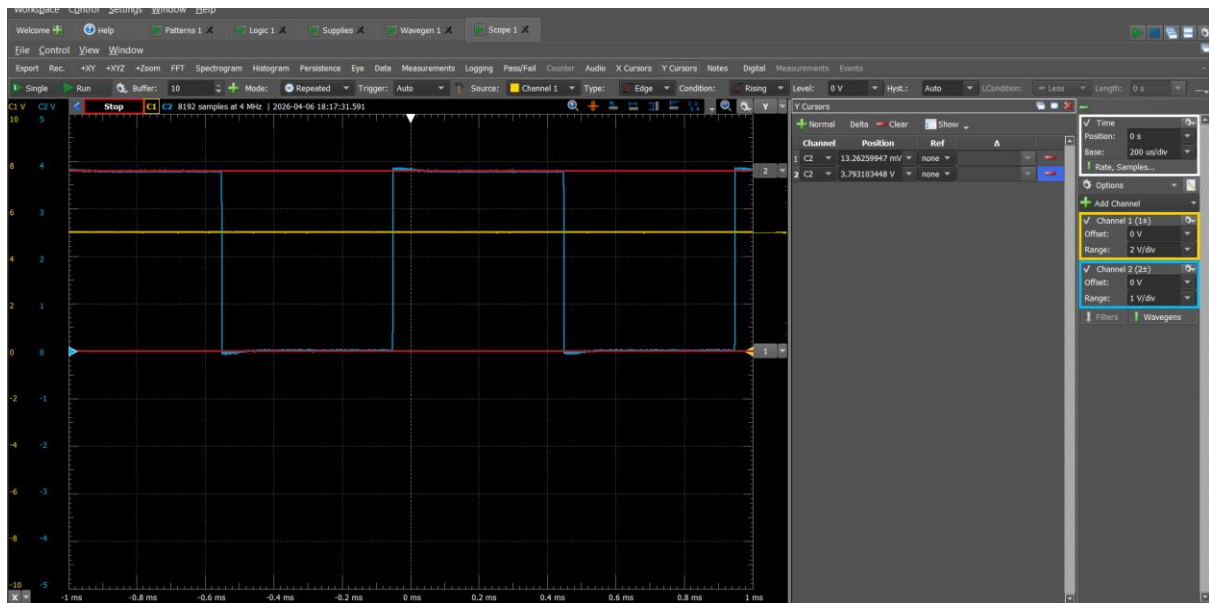


Figure 6: Visualization of logic high and low voltage when input A is fixed and input B is alternating

From the Fig above, when B is alternated between 0-5 V and A is kept high at 5V, the exact same output graph and logic state values from the previous graph are seen, with a high logic state (V_H) of 3.79V and low logic state (V_L) of 13.27 mV.

Hence the high and low logic states for the circuit can be defined as:

$$V_H = 3.79V \text{ and } V_L = 13.27mV$$

3. Timing

For the timing aspect of the design, input A was set to alternate between 0-5 V using a square wave, and input B set to 5V DC. A 100nF capacitor was then connected at the output to ground to simulate a load. Using the AD3s measurement tool the fall and rise times were found to be approximately 221.55 μ s and 199.05 μ s respectively.

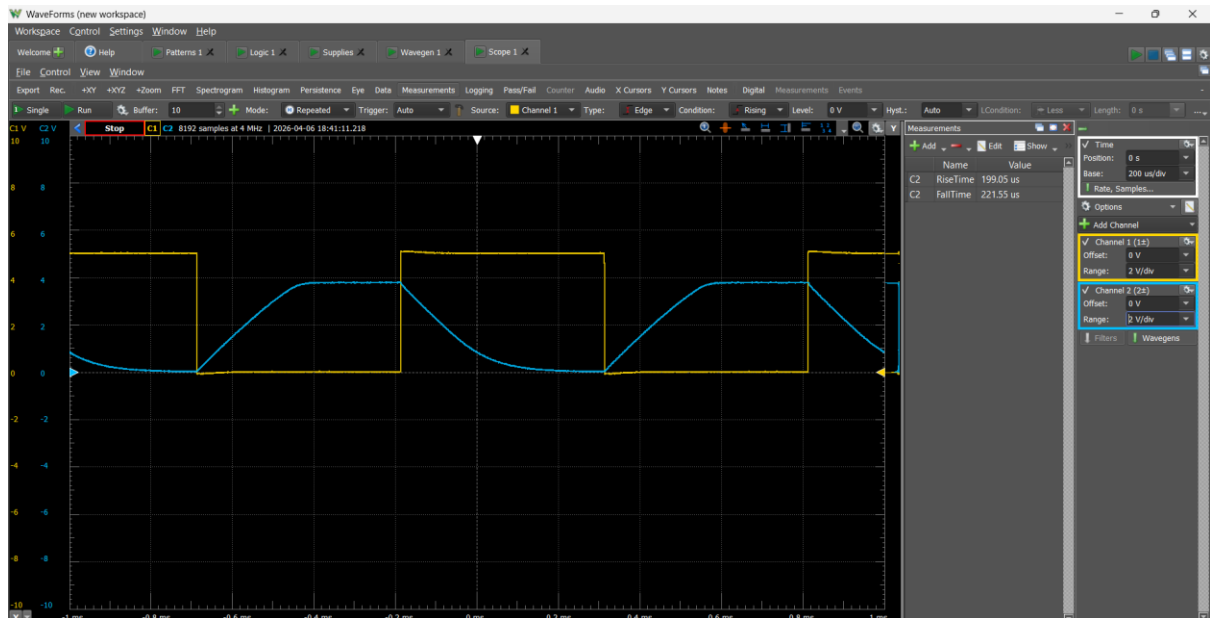


Figure 7: Propagation delay graph with rise and fall time measurements

Using the same graph, the propagation delay for the transition between high to low, and vice versa can be determined approximately by finding the time difference between the half of the square wave transition and half of the capacitors voltage transition. Since the square wave transitions almost instantaneously, the time difference between the instantaneous change and the halfway transition of the capacitor can be measured.

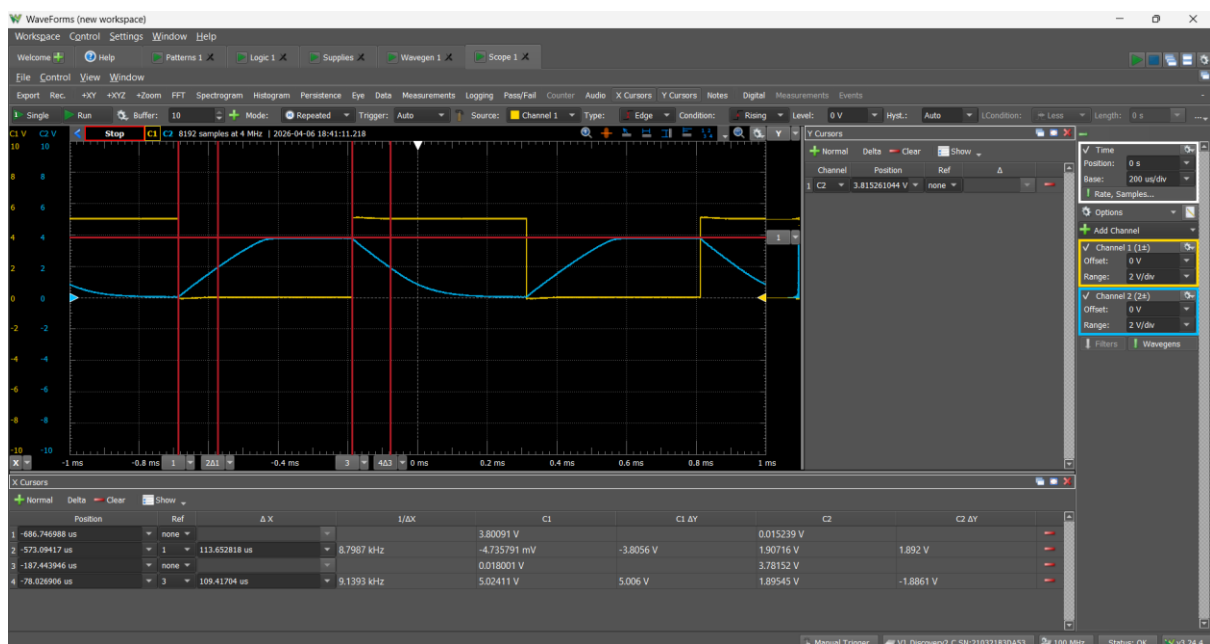


Figure 8: Propagation delay graph with measurements and cursors

From the figure above, the capacitor charges to a final voltage of approximately 3.815V. Half of this voltage would be approximately 1.9075V. The time values according to this values were measured and recoded. The low to high time delay and high to low time delay are listed below:

$$\tau_{pLH} \approx 113.653 \mu s$$

$$\tau_{pHL} \approx 109.417 \mu s$$

The average propagation delay can then be calculated by finding the average of the two:

$$\tau_p = \frac{\tau_{pLH} + \tau_{pHL}}{2} \approx \mathbf{111.535 \mu s}$$